

Comparison Of Unsupervised Machine Learning Algorithm For Dimensionality Reduction

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Abstract—The dimensionality of a dataset is the sum of all its variables, traits, and features. The objective is to minimize the number of columns used to depict this axis. Machine learning (ML) is a data-driven technique in which computers learn without human intervention by analyzing large amounts of historical data. Incredible uses of ML may be found in a variety of fields. Supervised learning, unsupervised learning, and reinforcement learning are the three primary learning challenges in ML. The authors of this work provide a thorough comparison of several approaches to Dimensionality Reduction. Training a model using an unlabeled dataset is known as unsupervised learning. The model picks up knowledge independently by studying the characteristics of the training dataset. The model then uses these learned characteristics to draw inferences about the test data. Clustering, k-means, agglomerative, principal component analysis, and Fuzzy C-means are just few of the many unsupervised learning methodologies and algorithms available. When using support vector machines with a radial basis function kernel for disease diagnosis based on gene or protein expression data, we present empirical comparison results comparing PCA to other dimension reduction (DR) methods (i.e. K-means Clustering Algorithm, Principal Component Analysis (PCA), Agglomerative, Apriori Algorithm, Fuzzy C-means). Among all this algorithms PCA gives the best results for the gene and protein dataset in terms of accuracy, cross validation rates and computing time. Our research shows that principle component analysis (PCA) not only outperforms other DR methods in terms of classification accuracy, but is also significantly more efficient since it does not need the optimization of a tuning parameter. The results lead us to believe that principal component analysis (PCA) might be a useful DR method for developing a trustworthy and unbiased classifier for gene or protein expression data.

Keywords—*Unsupervised Learning, Literature Review Survey, Machine Learning, Unsupervised Learning Approaches Clustering, PCA.*

I. INTRODUCTION

Machine learning techniques are being used to a wide range of bioinformatics problems at the moment, speeding up development in several areas of research [21, 25]. One of its primary focuses is the use of advanced statistical methods to microarray data, which contains the pattern of gene or protein expression of patients, in order to diagnose illness. Reducing the amount of features in a dataset is known as dimensionality reduction. For many common machine learning tasks like regression and classification, there simply aren't enough inputs. Features may be thought of as a synonym for these factors. It is possible to classify ML as a subsection of AI because of the ideas it borrows (AI). Fundamental to its ability to make forecasts are algorithms. With ML, computers may acquire knowledge without being expressly designed to do so, leading to smarter actions overall.

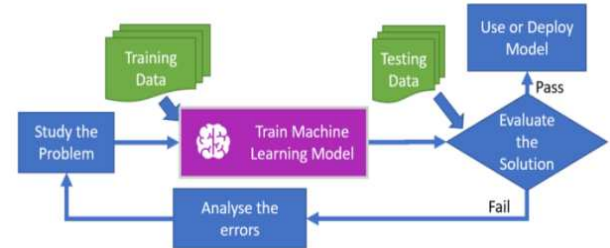


Fig. 1. Basic Architecture of ML

The above figure 1 shows the basic architecture of the ML algorithm. Supervised Learning, Unsupervised Learning, and Reinforcement Learning are the three primary sub-fields of ML. Supervised learning is the process of using a model trained on labelled data to generate predictions on unlabeled data. [3].

A. Unsupervised Learning

Algorithms for discovering patterns in unlabeled or unclassified data are known as "unsupervised learning." That's why we let the algorithms label and organize the data points inside the datasets on their own. The model may be used independently of human oversight [5].

In other words, the learning algorithm is not provided with any labels and must instead rely on its own inferences to discover the underlying structure in the incoming data. When there are no predetermined categories available, an AI system may nonetheless learn to classify data based on similarities and differences via a process called unsupervised learning [6,11]. The unsupervised learning can be categorized into clustering, association, autoencoders and anomaly detection as shown in the fig 2.

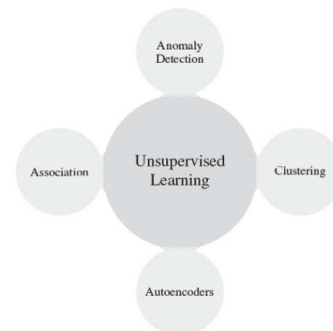


Fig. 2. ULT (Unsupervised Learning types)

1) Clustering

When it comes to unsupervised learning, the idea of clustering is crucial. The primary focus is on applying this method to unstructured data in order to discern meaningful patterns [13]. Training with no supervision. When applied to

your data, clustering algorithms will identify any existing natural clusters (groups). The number of clusters that your algorithms should find is also a variable. The granularity of these categories may be modified [7].

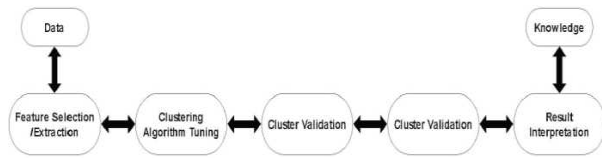


Fig. 3. Workflow of clustering in unsupervised ML

Clustering results may be graphically represented, as seen in figure 3. In order to organize the data, we divide the subjects into groups according to shared characteristics as in fig 4.

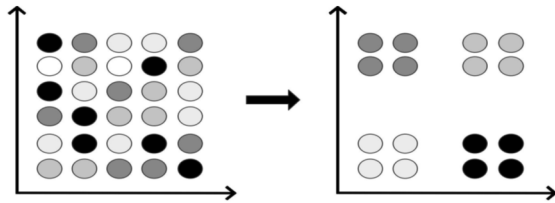


Fig. 4. Clustering

Clustering may be performed in a number of different ways, including by splitting the data, using a hierarchical structure, using overlap and probability as shown in fig 5. Partitioning is the process of organizing data into subsets, where each piece of information belongs to exactly one partition. Exclusive clustering is another name for it. Clustering methods that divide the data into smaller pieces are k-means. All the data is a cluster in Hierarchical clustering. The number of clusters is decreased by repeated unions of the two closest clusters. Data clustering using overlapping fuzzy sets. Each individual data point may have varying degrees of membership in two or more distinct groups. Here, a suitable membership value, such as Fuzzy C-Means, will be assigned to the data. Last but not least, probabilistic clustering employs the usage of probability distribution to produce the aforementioned subgroups [12].

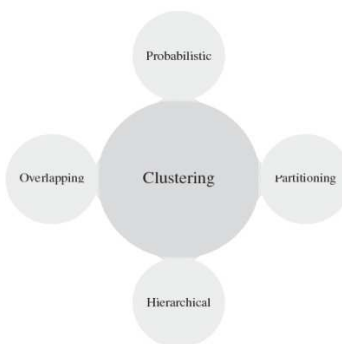


Fig. 5. Types of Clustering

2) Association

Because the data are not labelled, the algorithm in association rule is unsupervised learning, in which it attempts to learn on its own. In contrast to the prediction methods often employed to unearth interesting associations buried in vast datasets, association rules are only descriptive. This connection is often shown as a set of rules or a collection of frequently occurring items [8].

Whether it's a common pattern in a series of transactions or a newly discovered connection between two seemingly unrelated entities, association rules mining may help you find the gold in your data. Price bundling, assortment decisions, cross-selling, and customer clustering in retail all make use of this information. So does Market Basket Analysis (which products consumers like to buy together) and Market Share Analysis (how popular certain brands are among consumers). One way to look at it is as a more complex "what-if" scenario: if this, then that.

3) Anomaly Detection

Anomaly detection refers to any method that seeks for and identifies data set anomalies. These outliers might indicate abnormal network activity, the location of a faulty sensor, or the need to clear data before analysis. An anomaly is a deviation from or a departure from the typical distribution of data. If there is an unexpected spike in one direction of network traffic, for instance, it might indicate that a compromised machine is transmitting private data to an unauthorized server. Anomaly detection requires the ability to recognize or anticipate data points that differ from the norm [9]. The field of anomaly detection has several potential uses, including security (intrusion), finance (fraud), and security (military).

4) Auto encoders

Auto encoders are a kind of unsupervised learning that uses neural networks for the goal of representation learning. A feed forward neural network is an auto encoder when the input and output are the same. In order to replicate the result, they first take the input and compress it into a lower dimensional code. This "summary" or "compression" of the input is the code, which is also called the latent-space representation [10]. The autoencoder consists of the encoder, the code, and the decoder as shown in fig 6. The input is shrunk in size and a code is generated by the encoder; the decoder then utilises this code to recover the original input. An autoencoder requires an encoding technique, a decoding technique, and a loss function to evaluate the accuracy of the comparison between the encoded and decoded data. In its most basic form, auto encoders are a kind of compression technique that works by reducing the number of dimensions involved in a signal.



Fig. 6. Autoencoder Components

Locating and evaluating previous research on the subject at hand is the primary goal of this literature review. This literature evaluation will provide academics and practitioners with novel opportunities for further study. In this paper, we provide a high-level comparison of various Unsupervised Learning approaches and algorithms, as well as a detailed breakdown of the metrics used to evaluate the performance of each unsupervised learning model and an examination of how they stack up against one another in terms of accuracy.

B. Background

Developers of artificial intelligence, as well as data professionals tasked with visualizing and interpreting complicated data sets, might benefit from dimensionality reduction. It helps with data compression, which in turn makes the compressed data less in size and speeds up the processing of the data. Integrating cognitive thinking into a traditional

computer may aid in the resolution of issues involving pattern recognition, categorization, and forecasting by way of example mapping. These kind of models may be obtained using Artificial Neural Networks (ANN). Simply said, they are mathematical models representing a function, but they are coupled with a specific learning algorithm or rule to imitate human behavior.

C. Motivation

As a driving force, dimension reduction is useful for artificial intelligence (AI) engineers and data professionals that deal with large datasets, visualize and analyze complicated data, and do similar tasks. It helps with data compression, which in turn makes the compressed data less in size and speeds up the processing of the data. Learning may mean either gaining new information or improving existing information. Herbert Simon defines Machine Learning as the process of making adaptive modifications to a system so that it can do a set of tasks from the same population with more efficiency and effectiveness the following time around.

II. LITERATURE REVIEW

S. B. Kotsiantis et al. [1]. To forecast new occurrences, supervised machine learning seeks systems that can infer meaning from data provided by an outside source. Thus, in supervised learning, we want to develop a straightforward model of the dispersion of class labels in terms of predictor features. The resulting classifier is then used to assign class labels to testing examples for which both the predictor feature values and the class label value are known. This research looks at many approaches to supervised machine learning's categorization task. As it is unrealistic to expect a single article to provide a comprehensive review of all supervised machine learning classification algorithms (also called induction classification algorithms), we do hope that the references cited will cover the major theoretical issues, pointing the researcher in interesting research directions, and suggesting possible bias combinations that have not yet been explored.

Iqbal Muhammad et al. [2]. Training computers to learn from data or experience is central to the field of machine learning. Many useful uses of machine learning already exist, including as systems that analyze sales data to forecast client purchasing behavior, fraud detection, and email classifiers trained on messages to learn to discriminate between spam and non-spam. Although Supervised Learning, Unsupervised Learning, and Reinforcement Learning may all be used for association analysis in machine learning, this research will concentrate on the advantages and disadvantages of Supervised Learning Classification Algorithms.

Hastie et al. [4]. Statistics is always being tested by new challenges brought in by the scientific and business communities. These issues were often limited in scale in the beginning, stemming from trials in agriculture and industry. Statistical issues have become exponentially larger and more difficult to solve with the introduction of computers and the information age. Problems with statistics and computing in biology and medicine inspired the development of "bioinformatics," while "data mining" arose to meet the growing need for solutions to issues related to data storage, organization, and search. The role of the statistician, in an age when massive volumes of data are being created across many disciplines, is to make sense of this information by identifying meaningful trends and patterns and deducing "what the data says." Data-driven learning is the term for this process.

The author et al. [5] These days, machine learning algorithms are incredibly useful tools for medical diagnostics. Many challenging medical problems can be swiftly and readily solved utilizing machine learning algorithms and methodologies. Finding the root causes of an issue or accurately suggesting the best solution for it was challenging without these tools. Nearly all areas of medical science, including heart disease, blood sugar, blood transfusion, cancer prognosis, gender prediction, and many others, use ML techniques. In the medical and health sciences, both supervised and unsupervised machine learning algorithms are employed to identify the best course of therapy for every medical condition. The deployment of supervised machine learning techniques is carried out in this work for the data of Pregnant women are managed based on the type of delivery, such as a C-section or a regular delivery, using machine learning algorithms and methodologies. Doctors can take prophylactic measures to safeguard the health of both the expecting woman and the unborn child thanks to the research's categorization of the participants into caesarean and regular delivery instances.

The author et al. [6] Breast cancer is the major disease in the women, detecting this breast cancer at early stage will save any lives. The author proposes a model based on the PCA algorithm which gave accuracy, precision, sensitivity, and F-Measure as 97.53%, 97.59%, 95.75%, and 96.66%, respectively.

Pierre Baldi et al. [7]. Autoencoders are crucial in unsupervised learning and deep architectures for transfer learning and other applications. In spite of the critical function they provide, only linear auto encoders over the real numbers have been solved analytically. We provide a unifying mathematical framework for the investigation of autoencoders, both linear and non-linear. The most non-linear autoencoder, the Boolean autoencoder, may be treated analytically within this paradigm. The Boolean autoencoder's learning process may be seen of as an instance of a clustering problem, the solution to which is tractable in polynomial time for low numbers of clusters but intractable for high ones.

III. PROPOSED METHODOLOGY

In the proposed work as shown in fig 7, it is necessary to reduce the medical data, so a literature review is conducted. Based on the results of the review, we conclude that principal component analysis (PCA) is used for dimensionality reduction and the SVM algorithm is used for feature extraction. We also use encryption techniques to safeguard the data against hackers before sending it to the cloud for storage.

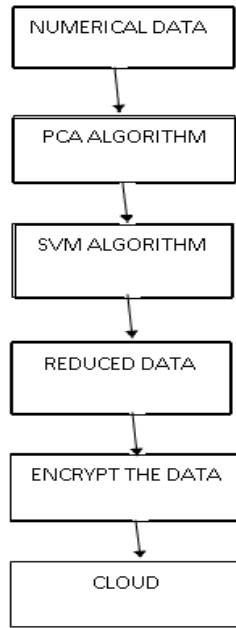


Fig. 7. Proposed block diagram

A. PCA for dimensionality reduction

Principal Component Analysis (PCA) is a well-known technique for automating the processes of feature extraction and dimensionality reduction by use of a linear transformation, saving time and eliminating the need for human oversight. Among the many well-known PCA applications in bioinformatics, the analysis of genomic data and gene expression levels stands out. Exploratory data analytics and signal de-noising in stock trading are two more common applications of principal component analysis.

Through analyzing the degree of correlation between characteristics, PCA allows us to discover previously unseen patterns in the data. In a word, principal component analysis (PCA) projects high-dimensional data into a new subspace with the same number of dimensions as the original data set or less dimensions in order to identify the directions of highest variation.

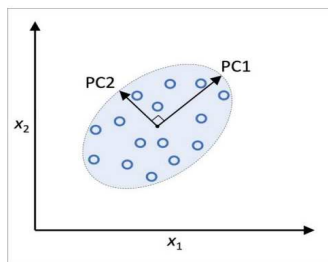


Fig. 8. PCA for dimensionality reduction

The original feature axes in the above fig 8 are labelled as x_1 and x_2 , while the main components are labelled as PC1 and PC2.

Using principal component analysis (PCA), we may transfer a sample vector (x) onto a new feature subspace (k) with less dimensions than the original feature space (d) for medical data, thereby reducing the number of dimensions.

IV. RESULT AND DISCUSSION

This section presents the findings and the discussion that follows. Each of the research questions that were asked and their answers may be found in the report. In this part, we discovered and explored a number of unsupervised learning methodologies and algorithms as shown in the fig 9. We have decided to use SVMs with an RBF kernel as our classifier. References [6, 9] provide an in-depth explanation of support vector machines. LIBSVM [8], a software package written in C++, was utilized for the tests. Except for Principal Component Analysis (PCA), the developers of the other four DR approaches (K-means Clustering Algorithm, Agglomerative, Apriori Algorithm, Fuzzy C-means) built and ran their own algorithms in MATLAB 7.0. Our testing was conducted on a P4 1.6GHz computer with 1GB of RAM.

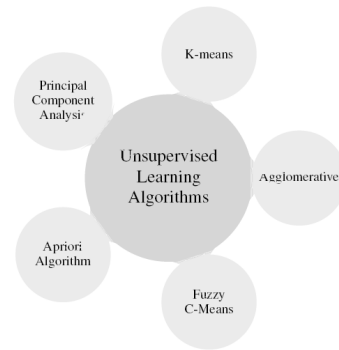


Fig. 9. Widely used Unsupervised Learning Algorithms

A. K-means Clustering Algorithm

As an Unsupervised Learning technique, K-Means Clustering is useful for analyzing large data sets and identifying patterns. It divides the unlabeled data set into K groups, where K is the total number of categories. As an example, if $K=2$, there will be two clusters, and if $K=5$, there will be five clusters [16,17].

This approach, which divides an unlabeled dataset into K groups, is iterative in nature. No two datasets may be placed in the same category of shared characteristics. It makes it possible to divide the information into several categories for easier collecting [23,24].

The pseudo-code for K-mean clustering is presented below:

Pseudo Code of K-means Clustering

1. You may specify the total number of clusters by picking a value "K."
2. Pick any K points or centers at random
3. Apply a centroid-based clustering algorithm to each data point and then divide the results into K clusters.
4. Compute the differences and re-enter the cluster centres.
5. Iterate through Step 3 until all data points have been reassigned to the clusters with the closest centroids.
6. Perform "Step 4" if and only if "any reassignment occurs."
7. Else: "End"

B. PCA

Principal component analysis (PCA) may be used to reduce the number of variables in a dataset that are interrelated. As a prerequisite, we must consult a distinct set of variables referred to as the major components (PCs). Principal component analysis (PCA) is a technique for reducing the dimensionality of the feature space by calculating the eigenvalue decomposition of an estimate of the covariance matrix of the data and then using the appropriate eigenvectors in the projection. A technique called "feature extraction" is implemented. Although it has many potential applications, at its core it is a statistical technique. Much of the development may be attributed to statisticians. In order to find new variables that are linear functions of the existing data points, we must preserve the existing variation. Principal component analysis attempts to do just that, by mapping high-dimensional data into a lower-dimensional plane. With principal component analysis, we can calculate how much variation there is in each characteristic. It is clear that there is a good separation between the classes when using a highly variable feature, which helps to minimize the dimensionality of the problem. Image processing and movie recommendation systems both benefit from PCA's usage. Utilizing PCA, only necessary characteristics are retained while the rest are discarded. Below is PCA's pseudo code:

1. Pseudo-Code of Principal Component Analysis (PCA)
2. Put the data into a coherent framework.
3. Data normalization must be performed.
4. Find Z's Covariance
5. The Eigen-Vectors and Eigen-Values must be calculated.
6. Rank the computed Eigen-Vectors in order.
7. Analyze the Key Elements and Recent Additions
8. To improve the new dataset, eliminate the irrelevant details.

C. Agglomerative

Each data point is treated as its own cluster in an agglomerative hierarchical clustering. Later, small groups are aggregated into larger ones. This strategy employs a bottom-up methodology. The clustering hierarchy is shown by a dendrogram [18,22].

The pseudocode of Agglomerative is as follows:

Pseudo-Code of Agglomerative

1. If the initial number of data points is K, then treat each data point as a separate cluster.
2. Combine the two neighboring data sets to build a larger cluster (K-1 clusters will be formed)
3. After that, combine the two nearest clusters to each other to generate new clusters (K-2 clusters will be formed)
4. Large clusters will be created indefinitely, or until K equals zero, whichever comes first.
5. Finally, the dendrograms are partitioned into several Clusters after being merged into a single large cluster.

D. Apriori Algorithm

In other words, the Apriori method is used to organize data into groups. To form correlations, the Apriori algorithm looks for patterns in frequently occurring data. Specifically, it operates on transaction databases. The strength of the connection between two things is established by the association rule. The item-set relationships are selected using a breadth-first search in this technique [19].

Pseudo-Code of the Apriori Algorithm

1. First, choose the minimal support and confidence among the item-sets and specify the support of item-sets in a transactional database.
2. Select supports with a greater support value rather than selecting supports with the lowest support value.
3. Find all the subset rules that have a greater confidence value than the stated threshold.
4. Sort the rules by weight from heaviest to lightest.

E. Fuzzy C-means

Clustering data using Fuzzy C-Means is an example of a gentle method for doing so. A score representing a data point's chance of being included in a certain cluster is awarded to it. Provides superior results than the k-means method [20,21] for overlapping data sets.

F. Data Sets

Table I provides a brief overview of the data sets we utilized, all of which can be found in [1]. Gene or protein expression levels for individuals are included in each dataset, along with metadata such as class labels for each individual record. The ALL-AML Leukemia data collection [14] includes 6817 genes from 72 people, and the gene expression data for each person is labelled to show whether they have Acute Lymphoblastic Leukemia (ALL) or Acute Myeloid Leukemia (AML) (AML). The Mixed-Lineage Leukemia (MLL) data collection [3] includes three subtypes of leukaemia: Acute Myeloid Leukemia (AML), Acute Lymphoblastic Leukemia (ALL), and M (MLL). Malignant pleural mesothelioma (MPM) and adenocarcinoma (ADCA) are the two types of lung cancer included in the Lung Cancer Data Set [15].

TABLE I. DESCRIPTION OF DATA SETS

Data Set	Data(Training Data + Test Data)	# Features	# Classes
ALL-AML Leukemia	72 (38+34)	7129	2
MLL Leukemia	72(57+15)	12582	3
Lung Cancer	181(32+149)	12533	2
DLBCL(Diffuse largeB-Cell Lymphoma)	47(32+15)	4029	2
DLBCL-Harvard	77(52+25)	6817	2
Ovarian Cancer	253(84+169)	6817	2

Using a single data point as a starting point, we may construct a similarity measure by traversing a manifold. Specifically, Isomap approximates geodesic distances by building a connectivity network from k-nearest neighbours and then computing similarities along the shortest path in the graph. In order to generate low-dimensional representations of such a nonlinearly produced proximity matrix, LLE builds a similar graph, but assigns edge weights by discovering the

optimal local convex/linear combinations of k-nearest neighbours to represent each original data point.

G. Implementation

Table II and Table III provide the original data sets' findings for CV rate and averaged test accuracy, respectively. It also shows the graphical representation of the same in the fig 10, 11. Table IV shows the findings with mislabeled data. It's important to keep in mind that the zero variations displayed in parenthesis in Tables III, IV are not the outcome of a single result but rather the average of several similar findings.

TABLE II. BEST CROSS-VALIDATION RATE ON TRAINING DATA

Algorithms	PCA	K-Means	Agglomerative	Apriori Algorithm	Fuzzy C-means
ALL-AML Leukemia	98.99	98.66	97.55	99.44	99.45
MLL- Leukemia	91.12	85.15	93.11	72.44	97.88
Lung Cancer	96.87	98.77	97.12	84.37	99.12
DLBCL	95.66	90.60	95.22	90.48	93.12
DLBCL- Harvard	97.55	96.38	99.12	81.45	99.11

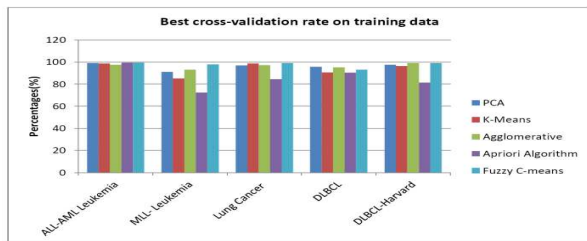


Fig. 10. Best cross-validation rate on training data

1) *Classification Accuracy*: Although DPDR often achieves above-average CV rates, sometimes it falls short. SVMs have more tuning options, which might be used to increase the CV rate. While Isomap and LLE provide respectable overall CV rates, they fail to sustain these results in test data. To put it another way, SVM retains its generalization capacity even without new parameters, but fine-tuned parameters in other DR approaches don't always provide the performance gain. Braga-Neto et al. [5] noted the significant variance of the cross-validation approach when used to estimate generalization error, which they said undermines its trustworthiness in cases of such disparity.

TABLE III. AVERAGE ACCURACY ON TEST DATA

Algorithms	PCA	K-Means	Agglomerative	Apriori Algorithm	Fuzzy C-means
ALL-AML Leukemia	98.60	89.75	92.15	77.13	77.12
MLL- Leukemia	99.52	67.11	91.52	73.33	92.33
Lung Cancer	88.12	81.18	86.15	59.12	74.55
DLBCL	82.15	74.12	73.44	73.45	71.55
DLBCL- Harvard	86.21	86.13	88.77	81.55	86.45

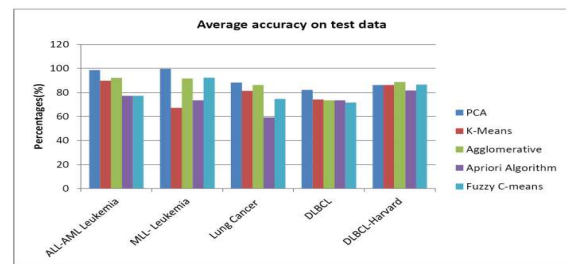


Fig. 11. Average accuracy on test data.

For both the ALL-AML and DLBCL data sets, LLE yielded subpar results due to its incorrect modelling of the underlying data structure.

In conclusion, DPDR's persistent success when used in tandem with nonlinear SVMs using the RBF kernel stands out.

2) *Robustness*: To evaluate the sturdiness of DR techniques, we conducted the trials detailed in Tables IV. The fig 12 is also computed to know the computing time for dimension reduction and cross validation.

The former protects against inaccurately labelled data, while the latter safeguards against the potential for measurement mistakes to add Gaussian noise. Since outliers, or incorrectly categorized data, are known to impair a classifier's performance, the findings are often worse than those in Table III. Although we used DR methods, our experimental findings indicated that performance varied widely over 5 trials per data set with different outlier selection, and in some cases it seemed to be much lower than 50%. Because SVM uses just support vectors to construct its own classifier model, it is very vulnerable to issues like the existence of outlier effects within the support vector pool, which may explain the steep drop in performance.

3) *Nonlinearity in Data*: SVM's ability to effectively handle any form of nonlinearly separable data is due to the nonlinear capabilities of classifiers with an appropriate kernel, such as the RBF kernel. Even while nonlinear data transformation is meant to help classifiers, it may instead impede them.

TABLE IV. COMPUTING TIMES (IN SECONDS) FOR DIMENSION REDUCTIONS AND CROSS-VALIDATIONS.

Algorithms	PCA	K-Means	Agglomerative	Apriori Algorithm	Fuzzy C-means
Dimension Reduction	0.182	48.203	594.55	75.66	188.75
Cross Validation	29	1650	1268	6087	6267
Total	28/6min	1667/29min	1897/33min	6155/107 min	6467/106min

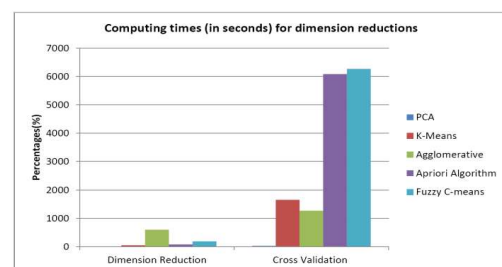


Fig. 12. Computing times (in seconds) for dimension reductions and cross-validations.

As evidence, consider Table II, where there is no clear winner when comparing linear versus nonlinear approaches. Then, we may achieve the same efficiency gains without resorting to NLDR's additional capabilities beyond just cutting down on the quantity of data we store. Similarity measures like Euclidean distance and cosine similarity are preserved in these reduced-dimensional representations, in accordance with the tenets of the SVM approach.

V. CONCLUSION

In this paper, Support vector machines with a radial basis function kernel are used for analyzing gene or protein expression data for the purpose of disease diagnosis, and we present empirical comparison results between PCA and other various dimension reduction (DR) methods (i.e. K-means Clustering Algorithm, Principal Component Analysis (PCA), Agglomerative, Apriori Algorithm, Fuzzy C-means). We find that principal component analysis (PCA) not only provides superior classification accuracy than competing DR approaches but also much higher efficiency, since it requires no optimization of a tuning parameter. From these findings, we infer that principal component analysis (PCA) is a potential DR technique for building a robust and distortion-free classifier for gene or protein expression data. This literature review will aid researchers in making decisions about which unsupervised learning strategy or algorithms to adopt for their own problem-solving. More databases, algorithms, and methods may be included in the future to provide even more precise direction. Multiple DR approaches for categorizing gene or protein expression data are compared. Principal component analysis (PCA) achieves not only the same findings as in the original space, but also considerably high efficiency both in creating lower dimensional representations and building classifiers, provided the classifier only uses Euclidean or cosine similarity measures. So in the proposed algorithm we will use PCA for reduction of the medical data and encrypt the data for protection against intruders and store the data onto cloud.

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